Smart Traffic Analysis & Congestion Prediction – Project Workflow

Step 1: Data Collection

Collected the Smart Traffic Management dataset from Kaggle.

The dataset contains traffic volume, vehicle counts, weather conditions, signal status, and location information.

Step 2: Data Preprocessing and Analysis

Loaded the dataset into Jupyter Notebook using Pandas.

Performed data cleaning and preprocessing.

Checked for missing values and duplicate records.

Converted the timestamp column into datetime format.

Created additional features such as:

Hour Column

Congestion Level Column (Low, Medium, High)

Step 3: Exploratory Data Analysis (EDA)

Analyzed traffic volume patterns.

Identified peak traffic hours.

Examined weather impact on traffic conditions.

Analyzed vehicle distribution and congestion levels.

Generated visualizations to understand traffic trends.

Step 4: Database Integration using PostgreSQL

Created a PostgreSQL database.

Imported the processed dataset into PostgreSQL.

Executed SQL queries to retrieve traffic insights.

Managed and organized traffic data efficiently.

Step 5: Power BI Dashboard Development

Connected PostgreSQL data to Power BI.

Developed an interactive Traffic Analysis Dashboard including:
- Traffic Volume KPI
- Average Vehicle Speed KPI
- Traffic Congestion Analysis
- Vehicle Type Breakdown
- Weather vs Traffic Analysis
- Traffic Location Map
- Traffic Details Table

Step 6: Machine Learning Model Development

Built a Linear Regression model using Scikit-Learn.

Selected the following features:
- Hour
- Temperature
- Humidity

Target Variable:
- Traffic Volume

Split the dataset into training and testing sets.

Trained and evaluated the model.

Step 7: Model Evaluation

Evaluated model performance using:

- R^2 Score
- Mean Absolute Error (MAE)

Generated traffic volume predictions.

Compared actual and predicted traffic values.

Step 8: Model Saving

Saved the trained machine learning model using Joblib.

Generated a .pkl file for future predictions without retraining.

Step 9: ML Prediction Dashboard

Exported prediction results to CSV format.

Imported prediction data into Power BI.

Created an ML Prediction Dashboard containing:

- Predicted Traffic Volume
- R^2 Score
- MAE
- Actual vs Predicted Traffic Scatter Plot

Step 10: Final Outcome

Successfully developed an end-to-end Smart Traffic Analysis and Congestion Prediction System.

Integrated Data Analysis, Database Management, Machine Learning, and Data Visualization into a single project.